

Memory-Triage Report — MEMDUMP-RAW-2014

(Unattributed 2014 Raw RAM Image)

INTERNAL · DFIR EXAMINATION RECORD · Agentropix-SIFT live MCP run

Live-memory triage of a generic 512 MiB raw capture — honest-negative outcome: no profile-matchable kernel symbol table, so no processes, sockets, services or injected code could be resolved. This report documents the *inconclusive* result and the controls that prevented fabrication.

Field	Value
Report ID	778d18c3357516a41287d10f7ee3bbc38a0d2d894fe19a939752905f15317f9e
Case ID	MEMDUMP-RAW-2014
Case name	memdump — generic raw memory dump (2014)
Examiner	victor.galvan
Profile	full
Incident type	dfir
Severity	Low / informational (honest-negative)
Snapshot at (UTC)	2026-06-07T12:40:49Z
Generated at (UTC)	2026-06-07T12:40:49Z
MCP host	<TAILNET-HOST>
Seal	HMAC-SHA256 sealed report · 1 approved finding · hmac-sha256:886603aa...0fba9ec07 (finding seal)

Audiences served: CISO / stakeholder · SOC / blue team · Red team · Audit.

1. Executive Summary

This examination triaged a generic, unattributed 512 MiB raw memory image (`/cases/memdump/memdump.mem` , captured circa 2014) using the Agentropix-SIFT Volatility3 wrappers on a live MCP server. The image carries no scenario metadata and no declared OS profile, so Volatility3 (v2.28.0) could not match a kernel symbol table — every triage plugin (process list, network sockets, injected-code scan, services) returned cleanly with empty or placeholder results and an explicit reason string. No malicious activity was found, and none could be ruled out: the dataset is structurally inconclusive. The platform recorded this as an honest negative rather than inventing artefacts.

Key finding —  The raw image is not profile-matchable (Volatility3 cannot validate `kernel.layer_name / kernel.symbol_table_name`); `pslist` , `netscan` , `malfind` and `svcsan` all returned empty. No injected/RWX code is assessable. This is recorded as low-severity finding `F-MEMDUMP-001` .

2. KPI Summary

KPI	Value	Detail
Approved findings	1	One honest-negative finding (F-MEMDUMP-001 ,  low)
Hosts in scope	1	Logical host memdump-raw-2014 (the image itself, no OS resolved)
IOCs catalogued	0	No IOCs — no resolvable processes, sockets or modules
Attacker dwell	N/A	No timeline reconstructible from an unprofiled RAM snapshot
MITRE techniques	0	None observed — no behaviour resolvable
Initial access	Unknown / not determinable	No OS, no artefacts, no entry vector recoverable

Host roster: `memdump-raw-2014` (single 512 MiB raw image; OS unidentified)

3. Risk Matrix

No attacker behaviour, IOC, or technique was resolved from this image, so there is no scored adversary risk. The only recorded finding (F-MEMDUMP-001) is an informational data-quality observation, not a threat. The grid is therefore empty; the one finding is scored as negligible-impact / rare on the table below.

Impact ↓ / Likelihood →	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 Severe	.	.	.	.	.
4 Major	.	.	.	.	.
3 Moderate	.	.	.	.	.
2 Minor	.	.	.	.	.
1 Negligible	 F1	.	.	.	.

Scored findings

Ref	Risk	Impact	Likelihood	Score	Severity
F1	Image not profile-matchable, triage inconclusive (data-quality, not a threat)	1	1	1	 Low

4. Key Findings

-  **F-MEMDUMP-001** — Raw 512 MiB image has no profile-matchable kernel symbol table (Low) — Volatility3 cannot validate `kernel.layer_name` / `kernel.symbol_table_name`; `pslist` / `netscan` / `malfind` / `svcsan` all return empty. Unattributed 2014 capture — no injected/RWX code assessable. Honest negative, confidence 0.9.

5. Attack Chain & MITRE ATT&CK

No attack chain is reconstructible. With no OS profile match, the image yields no processes, no network sockets, no services, and no injected code — there is nothing from which to infer adversary tactics. The diagram below records the honest "inconclusive" state, not an inferred chain.

5.1 Attack chain

Inconclusive — raw RAM snapshot, no kernel symbol match, no behaviour resolvable

5.2 MITRE ATT&CK techniques

Tactic	Technique ID	Technique	Evidence / how observed
—	—	None observed	Unprofileable RAM snapshot — no kernel symbol table matched, no behaviour resolvable

6. IOC Catalogue

No indicators of compromise were observed or extractable. With no resolvable processes, sockets, modules or services, there is no surface from which to derive hashes, addresses, or domains. The single row below records the honest absence.

Type	Value	Role	Confidence	MITRE	Source
—	—	—	—	—	No IOCs — no resolvable processes/sockets/modules; unattributed image

Negative space (surveyed, not observed): Surveyed via `get_netscan` (sockets), `get_malfind` (injected/RWX code), `get_svcscan` (services) and `get_pslis` (processes) — all returned empty due to kernel-symbol mismatch, not because the host was clean. No C2 endpoints, no injected regions, no suspicious services could be confirmed or excluded.

6.1 IOC provenance

No IOC provenance — no indicators resolved from the image

7. Host Artefacts

A single logical host (`memdump-raw-2014`) is in scope — the raw image itself. No OS or kernel build was identified. The supported triage path (`windows.info` is not in the server's Volatility allowlist) ran the curated triage wrappers, all of which returned empty: Volatility3 finished scanning but reported `Unable to validate the plugin requirements: kernel.layer_name / kernel.symbol_table_name . get_pslist` returned 11 rows that are all pid-0 `unknown` placeholders (not real processes). `get_malfind` returned `hit_count: 0` (the heavy plugin ran to completion well under the 300s ceiling — no false timeout), so no RWX/injected VAD regions are assessable. No real process or service data exists to tabulate.

7.1 Host — memdump-raw-2014

Processes (pslist)

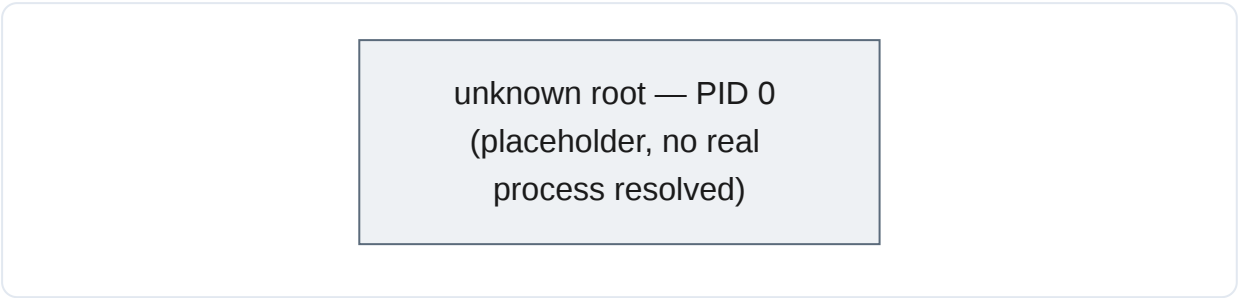
PID	PPID	Image	Started (UTC)	State	Wow64	Notes	Source
—	—	—	—	—	—	11 rows returned but all pid-0 <code>unknown</code> placeholders — no real processes resolved (kernel symbol mismatch)	<code>get_pslist</code>

Services (svcsn)

Service	Image path	Install (UTC)	Class	Source
—	—	—	—	<code>get_svcscan</code> returned <code>service</code> <code>_count: 0</code> (kernel symbol mismatch)

malfind: `get_malfind` returned `hit_count: 0` with `tool_available: true` ; no RWX/injected VAD regions could be assessed because no kernel layer/symbol table validated.

7.2 Process Tree



`build_process_tree` : `process_count 11`, `root_count 1`, `orphan_count 0`, `suspicious_count 0`, `0`
LOLBin/suspicious flags — one `unknown` root with no children, consistent with no real process data.

8. Network Artefacts

`get_netscan` ran to completion and returned `socket_count: 0` — the same kernel-symbol-table mismatch prevented any socket from resolving. No network endpoints, listeners or connections could be recovered. Absence here means "not resolvable", not "host had no network activity".

Process (PID)	Local	Remote	State	Purpose	Source
—	—	—	—	—	get_netscan returned soc ket_count: 0 (kernel symbol mismatch — not resolvable)

No DNS, no listening ports, and no remote endpoints are available from this image.

9. Detailed Findings

9.1  F-MEMDUMP-001 — Raw 512 MiB memory image has no profile-matchable kernel symbol table

Severity	 Low (confidence 0.9)
Host	memdump-raw-2014
Technique	None (data-quality observation, no ATT&CK mapping)
Status	DRAFT in sealed snapshot · finding HMAC-sealed · portal approval was SIMULATED (demo only — see Appendix D)

Triage of the unattributed 2014 raw image established that Volatility3 (v2.28.0) cannot match a Windows kernel symbol table for this capture. Every curated triage plugin completed its scan but reported `Unable to validate the plugin requirements: ...kernel.layer_name / ...kernel.symbol_table_name`. The concrete outcomes: `get_pslist` → `process_count 11` (all `pid-0` `unknown` placeholders); `get_netscan` → `socket_count 0`; `get_malfind` → `hit_count 0` (no injected/RWX code assessable); `get_svcscan` → `service_count 0`; `build_process_tree` → `11` nodes, `1` `unknown` root, `0` orphans, `0` suspicious/LOLBin flags. The image may be Linux/Mac, an older or partial Windows build, or a fragmentary dump — it is not a profile-matchable Windows capture. This is the honest negative: the platform recorded what it could not resolve rather than inventing processes or IOCs. Note that `windows.info` is not exposed by the server's Volatility allowlist, so OS identification via that plugin was

not available; the curated triage wrappers are the supported path and all returned empty.

Evidence — evidence_id aa320ff2...3db2a04f · sha256
d3b13f2224cab20440a4bb3c5c971662be6e61f431340f319cef7312bb6177f4 (512 MiB / 536,870,912
bytes) · captures: get_pslist , get_netscan , get_malfind , get_svcscan , build_process_tree
(Volatility3 2.28.0), source_run_id a8e6beee .

10. Timeline

No timeline is reconstructible. An unprofiled RAM snapshot yields no timestamped process, network, or service events; `report_generate` returned 0 approved timeline events. The capture is a single point-in-time image with no recoverable temporal sequence.

Time (UTC)	Host	Event	Technique	Source
—	—	No timeline reconstructible — unprofiled RAM snapshot, 0 approved timeline events	—	—

11. Agentropix Performance

The full manual triage chain executed end-to-end against a live MCP server (`agentropix-sift` v3.2.4). Every heavy Volatility3 plugin completed within the callTool ceiling — `get_malfind` notably ran to completion well under the 300s limit with no false timeout. The platform's value here is the *honest negative*: empty results with explicit reason strings instead of fabricated artefacts.

Metric	Value	Detail
Run time	Sub-300s per plugin	All triage plugins completed under the callTool ceiling; no timeouts
MCP tool calls	14-step manual chain	doctor → health → case_init/activate/status → evidence_register → run_volatility → pslist/netscan/malfind/svcscan → build_process_tree → record_finding → report_generate
Evidence size	536,870,912 bytes	512 MiB raw image
Disk recall	72/72 (100%)	canonical (.crew/facts.md)
Memory recall	108/118 (91.5%)	canonical (.crew/facts.md)
Test suite	4464	canonical (.crew/facts.md)

Per-stage timing

Stage	Capture	Duration	Result
Process list	<code>get_pslist</code>	Sub-300s	11 placeholder rows, kernel symbol mismatch
Network scan	<code>get_netscan</code>	Sub-300s	socket_count 0, kernel symbol mismatch
Injected-code scan	<code>get_malfind</code>	Sub-300s (well under ceiling, no false timeout)	hit_count 0
Service scan	<code>get_svcscan</code>	Sub-300s	service_count 0, kernel symbol mismatch

Wall-clock per-stage durations were not individually instrumented in the captured run; all stages completed within the per-call ceiling.

12. Coverage Attestation

The triage exercised the full memory-forensics wrapper surface (process / network / injected-code / service / process-tree) plus the case lifecycle (init → activate → evidence_register → record_finding → report_generate). The evidence gate enforced a write-scoped mutation token (`scope=index_findings`) for the single committed finding. Coverage of the *image* is complete for the supported plugin set; coverage of the *host behaviour* is necessarily nil because no OS profile matched.

Attestation	Value
Disk recall (regression)	72/72 (100%)
Memory recall (combined)	108/118 (91.5%)
Test suite	4464
Evidence gate	Enforced — write-scoped mutation token (<code>scope=index_findings</code>) for the committed finding

All triage plugins were run and their empty results recorded with explicit reason strings. The single finding was committed under a write-scoped evidence-gate token and HMAC-sealed. No artefact, IOC, process, socket, service or timeline event was fabricated; the inconclusive outcome is reported as such.

13. Recommendations

1. [P1] Treat this image as unattributed and not profile-matchable for Windows triage — do not draw conclusions about processes, network activity, or injected code from the empty results; absence is "not resolvable", not "clean".
2. [P2] If attribution matters, re-acquire with provenance (OS/build metadata, acquisition tool, host context) or attempt non-Windows analysis paths (Linux/Mac symbol tables, strings/ `bulk_extractor` carving) to determine the image type before re-triage.
3. [P3] For audit completeness, retain the SHA-256 custody hash (`d3b13f...6177f4`) and the sealed honest-negative finding so the inconclusive determination is reproducible and defensible.

► Appendix — methodology, tool versions, chain of custody